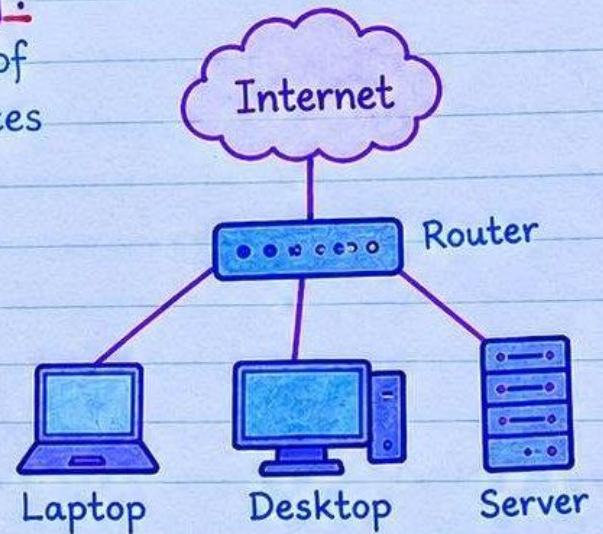


Computer Networking

1. What is Computer Networking?

Computer networking is the process of connecting computers and other devices together so they can communicate, share resources and exchange data with each other. It allows devices to work together over a network (either small like a home network or large like the internet).



2. Types of Networks

Networks can be classified based on their size and scope:

- PAN – Personal Area Network (e.g. Bluetooth, phone hotspot).
- LAN – Local Area Network (e.g. home, office, school).
- MAN – Metropolitan Area Network (e.g. city wide).
- WAN – Wide Area Network (e.g. countries, internet).

3. Key Devices in a Network

- Router – Connects different networks and routes data.
- Switch – Connects devices within the same network.
- Hub – Sends data to all devices (less common).
- Modem – Connects your network to the internet.
- Access Point – Provides wireless network (WiFi).

4. Common Network Types

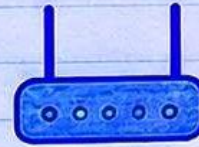
- Wired Network – Uses cables (Ethernet).
- Wireless Network – Uses radio waves (WiFi).
- VPN – Creates a secure connection over the internet.

Computer Networking (Page 2)

5. Network Devices

Network devices help connect, manage and route data between devices and networks.

- Router - Routes data between different networks.
- Switch - Connects devices in the same network.
- Hub - Sends data to all devices (less common).
- Modem - Connects a local network to the ISP.
- Access Point - Provides WiFi access to devices.



Router



Switch



Modem

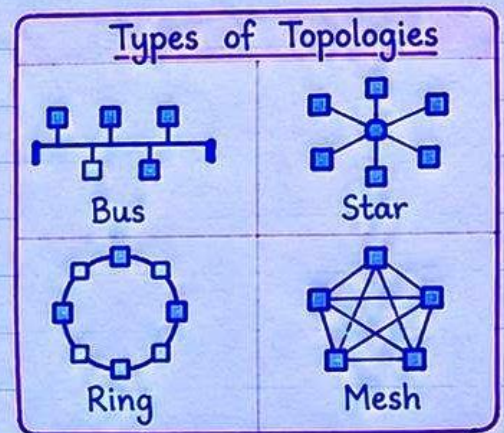


Access Point

6. Network Topologies

Network topology is the arrangement of devices in a network.

- Bus - All devices share one main cable.
- Star - All devices connect to a central device.
- Ring - Devices are connected in a circle.
- Mesh - Devices connect to many other devices.



7. IP Addressing

IP address is a unique number given to each device on a network.

- IPv4 - 32-bit address (e.g. 192.168.1.1).
- IPv6 - 128-bit address (e.g. 2001:0db8::1).

IP addresses can be:

- Public - Used on the internet.
- Private - Used in local networks.

Examples of IP Ranges

Private (IPv4):

192.168.0.0 - 192.168.255.255

Loopback (IPv4):

127.0.0.1

Public (IPv4):

1.0.0.0 - 223.255.255.255

8. Network Protocols

Protocols are rules that control how data is sent, received and understood.

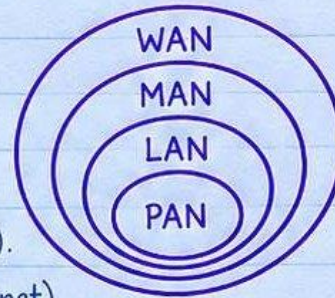
- TCP - Reliable, connection-oriented (e.g. web, email).
- UDP - Faster, connectionless (e.g. video streaming, online games).
- HTTP - Transfers web pages.
- HTTPS - Secure version of HTTP (uses encryption).
- FTP - Transfers files over a network.

Computer Networking (Page 3)

9. Network Types

Networks can be classified based on their size, connection and purpose.

- **PAN** - Personal Area Network (e.g. Bluetooth).
- **LAN** - Local Area Network (e.g. home, office).
- **MAN** - Metropolitan Area Network (e.g. city wide).
- **WAN** - Wide Area Network (e.g. countries, internet).

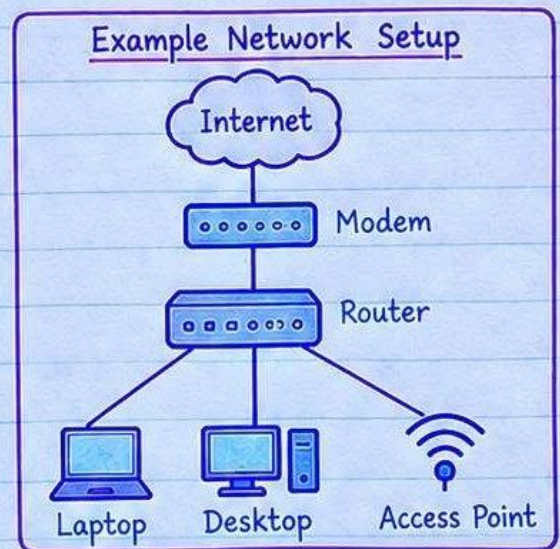


Larger networks cover bigger areas.

10. Network Devices

Network devices are used to connect, manage and control the flow of data in a network.

- **Router** - Routes data between networks.
- **Switch** - Connects devices in the same network.
- **Hub** - Sends data to all devices.
- **Modem** - Connects to the ISP (internet).
- **Access Point** - Provides WiFi access.



11. Network Cables

Network cables are used to connect devices in a network.

- **Twisted Pair** - Used in LAN (e.g. Cat5e, Cat6).
- **Coaxial Cable** - Used in older networks (less common).
- **Fiber Optic** - Used for high speed and long distance.

Types of Connectors



RJ45
(Ethernet)



BNC
(Coaxial)



SC
(Fiber Optic)

12. IP Address Classes

IP addresses are divided into classes to make address management easier.

Class	Range (First Octet)	Default Subnet Mask	Used For
A	1 - 126	255.0.0.0	Large networks
B	128 - 191	255.255.0.0	Medium networks
C	192 - 223	255.255.255.0	Small networks
D	224 - 239	—	Multicast
E	240 - 255	—	Experimental

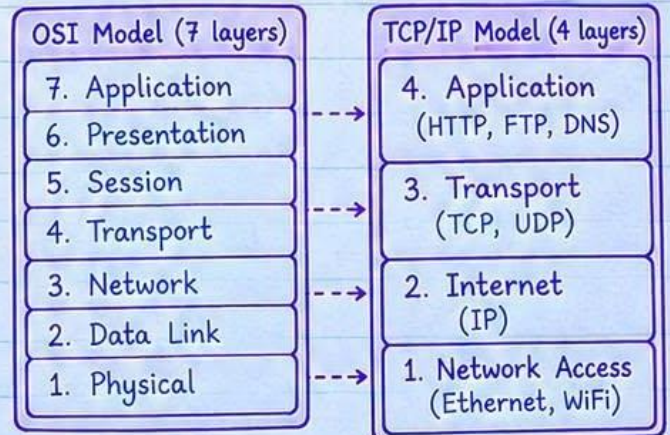
★ Private IP ranges: 10.0.0.0 - 10.255.255.255 172.16.0.0 - 172.31.255.255 192.168.0.0 - 192.168.255.255

Computer Networking (Page 4)

13. OSI & TCP/IP Models

The OSI and TCP/IP models are frameworks used to understand how data moves over a network.

- OSI Model – 7 layers (theoretical model).
- TCP/IP Model – 4 layers (practical model).



14. Differences Between Routing and Switching

Both routing and switching help data move in a network, but they work at different layers.

Feature	Routing	Switching
Layer	Network layer (Layer 3)	Data link layer (Layer 2)
Purpose	Connects different networks	Connects devices in the same network
Device	Router	Switch
Uses	IP addresses	MAC addresses
Example	RIP, OSPF, BGP	VLAN, STP

15. IP Types

IP addresses can be classified into different types based on their usage and structure.

- IPv4 – 32-bit address (e.g. 192.168.1.1).
- IPv6 – 128-bit address (e.g. 2001:0db8::1).
- Public IP – Accessible from anywhere on the internet.
- Private IP – Used within a local network (not routable on the internet).

Private IP Ranges

10.0.0.0 – 10.255.255.255
 172.16.0.0 – 172.31.255.255
 192.168.0.0 – 192.168.255.255

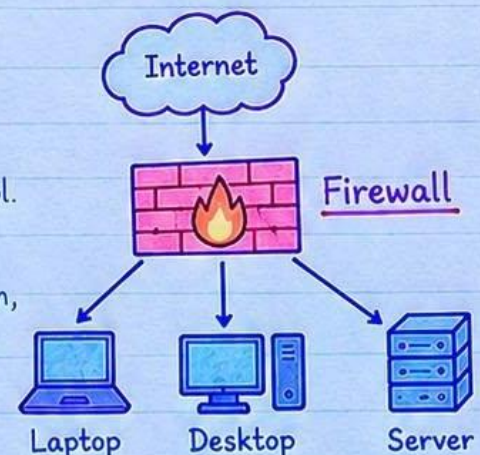
Public IP Range

1.0.0.0 – 223.255.255.255
 (excluding private ranges)

16. Firewall Types

A firewall is a security device or software that monitors and controls incoming and outgoing network traffic based on security rules.

- Packet Filtering – Filters data based on IP, port, protocol.
- Stateful Inspection – Tracks connection state.
- Next-Generation Firewall (NGFW) – Deep packet inspection, advanced security features.
- Proxy Firewall – Acts as an intermediary between networks.

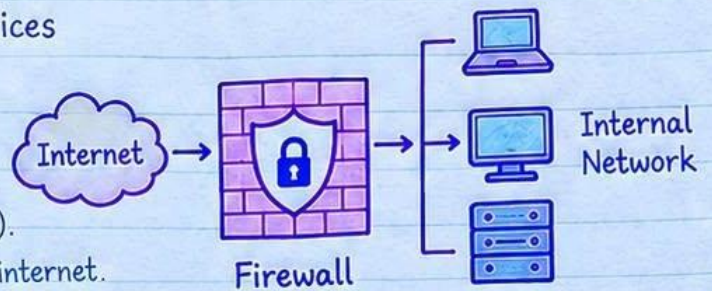


Computer Networking (Page 5)

17. Network Security

Network security protects data and devices from threats and unauthorized access.

- Firewall - Blocks unwanted traffic.
- Antivirus - Detects and removes malware.
- Encryption - Keeps data safe (e.g. HTTPS).
- VPN - Creates a secure tunnel over the internet.
- IDS/IPS - Detects and prevents attacks.



18. Network Devices (Detailed)

Network devices help in connecting, managing and controlling the network.

Device	Function	Example
<u>Router</u>	Routes data between networks.	Home router, ISP router
<u>Switch</u>	Connects devices in the same network.	24-port switch
<u>Hub</u>	Broadcasts data to all devices.	Old network hub
<u>Modem</u>	Connects to the ISP (internet).	Cable modem
<u>Access Point</u>	Provides WiFi access.	WiFi router
<u>Repeater</u>	Extends WiFi range.	WiFi extender

19. Network Models (Quick Recap)

Two common models are the OSI and TCP/IP models.

OSI Model (7 layers)
7. Application
6. Presentation
5. Session
4. Transport
3. Network
2. Data Link
1. Physical

TCP/IP Model (4 layers)
4. Application (HTTP, FTP, DNS)
3. Transport (TCP, UDP)
2. Internet (IP)
1. Network Access (Ethernet, WiFi)

Key Difference:

- OSI has 7 layers (theoretical).
- TCP/IP has 4 layers (practical).
- TCP/IP is widely used on the internet.
- OSI helps in understanding network concepts.

20. Common Network Commands

These commands are used to troubleshoot and manage networks.

- ping - Checks if a device is reachable.
- ipconfig - Shows IP configuration (Windows).
- ifconfig - Shows IP configuration (Linux).
- tracert - Shows the route to a destination.
- netstat - Shows active connections.
- nslookup - Looks up domain names.
- arp - Shows ARP table.
- route - Shows routing table.

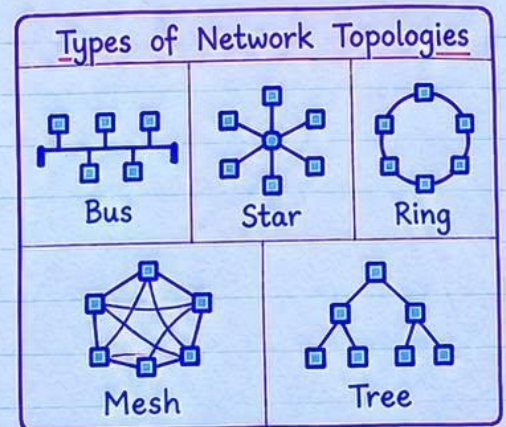


Computer Networking (Page 6)

21. Network Topologies (Types)

Network topologies define how devices are connected in a network.

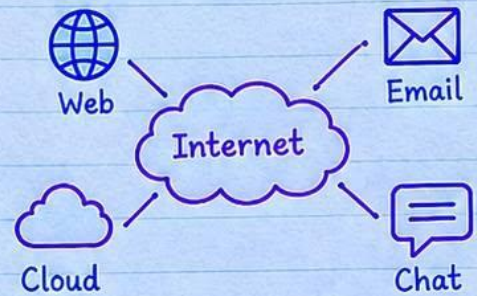
- **Bus** - All devices share one main cable.
- **Star** - All devices connect to a central device.
- **Ring** - Devices connect in a circle.
- **Mesh** - Devices connect to many other devices.
- **Tree** - Hierarchical connection (like a tree).



22. Internet & Its Services

The internet is a global network of networks that connects millions of devices and provides many services.

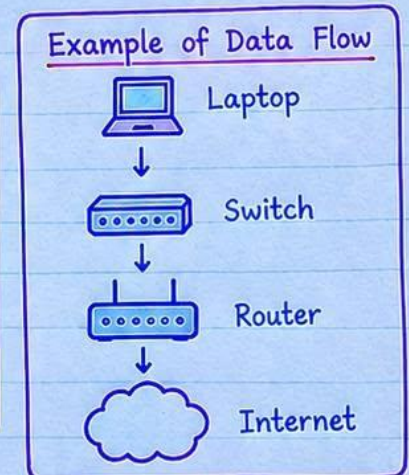
- **Web** - Access to websites (e.g. Google, YouTube).
- **Email** - Send and receive messages.
- **FTP** - Transfer files.
- **Chat** - Real-time communication.
- **Cloud** - Store data online (e.g. Google Drive).



23. Network Devices and Their Functions

Each device plays a role in moving data across a network.

Device	Function
Router	Routes data between networks.
Switch	Connects devices in the same network.
Hub	Broadcasts data to all devices.
Modem	Connects to the ISP (internet).
Access Point	Provides WiFi access.
Repeater	Extends WiFi range.



24. Conclusion

Computer networking helps devices and people communicate, share resources and access information. It uses different devices, protocols and topologies to make the internet and local networks work efficiently.

Stay Connected
Stay Learning
Keep Coding 😊